

Akachukwu D. Obi, Ph.D.

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Professional Appointments

Assistant Professor of Integrated Sciences: Chemistry of Materials <i>Claremont McKenna College (CMC)</i>	2026 - present
Postdoctoral Associate <i>Massachusetts Institute of Technology (MIT)</i> Advisor: Prof. Alex Radosevich	2025 - 2026
Postdoctoral Impact Fellow <i>MIT Climate and Sustainability Consortium (MCSC)</i>	2023 - 2025
Science Teaching Faculty <i>Tabor Academy</i>	2017 - 2018
Math Teaching Fellow <i>St. Paul's School (SPS)</i>	2015 - 2017
Summer Teaching Faculty <i>Phillips Andover Academy</i>	2015 - 2017

Education

Ph.D., Chemistry <i>University of Virginia (UVA)</i> Thesis: Electronically Flexible Carbon Donors in Magnesium and Bismuth Chemistry Advisor: Prof. Robert J. Gilliard Jr.	2018 - 2022
M.S., Education <i>University of Pennsylvania (PennGSE)</i> Thesis: How can I help my students build math confidence?	2015 - 2017
B.S., Chemistry <i>Bates College</i> Thesis: Nonlinear Optical Behavior in Plasmonic Gold Nanoparticles Advisor: Prof. Matthew J. Côté	2011 - 2015

Publications

† equal contribution, ‡ mentee, * corresponding author

Journal Articles

- (21) Kuo, F.-Y.; Byun, G. H.; **Obi, A.D.***; Junor, G.P.*; Gallant, B.M.* [Oxygen-tolerant electrochemical CO₂ separation using N-heterocyclic imines with superstoichiometric release per electron](#). *Nature Energy* **2026**, 1–9, [Preprint available](#) in ChemRxiv.
- (20) McMillion, N. D.; Frey, N. C.; Müller, P.; **Obi, A.D.***; Wilson, D.J.D.*; Gilliard Jr., R.J.* [Carbene-Stabilized Monomeric Cyanogen N-Oxide](#). *Journal of the American Chemical Society* **2026**, 148, 36071–36082.
- (19) **Obi, A.D.***; Moore, E. A.; Chun, J.; Olivetti, E. A. How will carbon capture and storage scale in hard to abate industries?. **2026**, Submitted.
- (18) **Obi, A.D.†***; Olivetti, E. A.; Junor G.†* Rethinking Carbon Capture Chemical Design to Address Critical Challenges in Deployment for Net-Zero. **2026**, Submitted.

- (17) Stopper Jr., M.J.[†]; **Obi, A.D.**[†]; Machost, H.R.[†]; McMillion, N. D.; Molino, A.; Cook, E. N.; Nichols, A. W.; Freeman, L. A.; Stegner, S. G.; Dickie, D. A.; Wilson, D. J.; Machan, C. W.; Jr., R. J. G. **Isolation and Reactivity of Carbene-Stabilized Carbon Disulfide Radical Anions.** *Inorganic Chemistry* **2025**, *64*, 3875–3884.
- (16) Deng, C.-L.; **Obi, A. D.**; Tra, B. Y. E.; Sarkar, S. K.; Dickie, D. A.; Gilliard Jr, R. J. **Air-and photo-stable luminescent carbodicarbene-azaboracene ions.** *Nature Chemistry* **2024**, *16*, 437–445, Highlighted in **MIT News**.
- (15) **Obi, A. D.**; Deng, C.-L.; Alexis, A.J.H.[‡]; Dickie, D. A.; Gilliard Jr, R. J. **Geminal Bimetallic Coordination of a Carbene to Main-Group and Transition Metals.** *Chemical Communications* **2024**, *60*, 1880–1883, **Hot Paper**.
- (14) Barker, J. E.; **Obi, A. D.**; Dickie, D. A.; Gilliard Jr, R. J. **Boron-Doped Pentacenes: Isolation of Crystalline 5, 12-and 5, 7-Diboratapentacene Dianions.** *Journal of the American Chemical Society* **2023**, *145*, 2028–2034.
- (13) Sarkar, S. K.; Hollister, K. K.; Molino, A.; **Obi, A. D.**; Deng, C.-L.; Tra, B. Y. E.; Stewart, B. M.; Dickie, D. A.; Wilson, D. J.; Gilliard Jr, R. J. **Bis (9-Boraphenanthrene) and Its Stable Biradical.** *Journal of the American Chemical Society* **2023**, *145*, 21475–21482, Highlighted in **Synfacts 2024**, *20*, 0022 and **ChemistryViews**.
- (12) **Obi, A. D.**; Dickie, D. A.; Tiznado, W.; Frenking, G.; Pan, S.; Gilliard Jr, R. J. **A Multidimensional Approach to Carbodiphosphorane–Bismuth Coordination Chemistry: Cationization, Redox-Flexibility, and Stabilization of a Crystalline Bismuth Hydridoborate.** *Inorganic Chemistry* **2022**, *61*, 19452–19462, **Editor's Choice Article. Most Read List**.
- (11) **Obi, A. D.**; Freeman, L. A.; Coates, S.J.[‡]; Alexis, A.J.H.[‡]; Frey, N. C.; Dickie, D. A.; Webster, C. E.; Gilliard Jr, R. J. **Carbene–Calcium Silylamides and Amidoboranes.** *Organometallics* **2022**, *41*, 3064–3072, **Most Read List**.
- (10) **Obi, A. D.**; Frey, N. C.; Dickie, D. A.; Webster, C. E.; Gilliard Jr, R. J. **N-Heterocyclic Carbene-Assisted Reversible Migratory Coupling of Aminoborane at Magnesium.** *Angewandte Chemie International Edition* **2022**, *61*, e202211496.
- (9) Freeman, L. A.; **Obi, A. D.**; Machost, H. R.; Molino, A.; Nichols, A. W.; Dickie, D. A.; Wilson, D. J.; Machan, C. W.; Gilliard, R. J. **Soluble, crystalline, and thermally stable alkali CO₂[−] and carbonite (CO₂^{2−}) clusters supported by cyclic (alkyl)(amino) carbenes.** *Chemical Science* **2021**, *12*, 3544–3550, **Invited Special Issue: Celebrating 10 years of Chemical Science. Preprint in ChemRxiv.** Highlighted in **C&EN News**.
- (8) Godbold, P.; Johnson, G.; **Obi, A. D.**; Brown, R.; Hwang, S.; Gilliard Jr, R. J.; Zhang, S. **Surfactant removal for colloidal nanocrystal catalysts mediated by N-heterocyclic carbenes.** *Journal of the American Chemical Society* **2021**, *143*, 2644–2648, Highlighted in **ChemistryViews**.
- (7) **Obi, A. D.**; Machost, H.R.[‡]; Dickie, D. A.; Gilliard Jr, R. J. **A Thermally Stable Magnesium Phosphaethynolate Grignard Complex.** *Inorganic Chemistry* **2021**, *60*, 12481–12488.
- (6) **Obi, A. D.**; Freeman, L. A.; Dickie, D. A.; Gilliard Jr, R. J. **N-Heterocyclic Carbene-Mediated Ring Opening of Reduced Diazamagnesiums.** *Organometallics* **2020**, *39*, 4575–4583.
- (5) **Obi, A. D.**; Walley, J. E.; Frey, N. C.; Wong, Y. O.; Dickie, D. A.; Webster, C. E.; Gilliard Jr, R. J. **Tris (carbene) Stabilization of Monomeric Magnesium Cations: A Neutral, Nontethered Ligand Approach.** *Organometallics* **2020**, *39*, 4329–4339, **Invited Special Issue: Organometallic Chemistry of the Main Group Elements**.
- (4) Freeman, L. A.; Walley, J. E.; **Obi, A. D.**; Wang, G.; Dickie, D. A.; Molino, A.; Wilson, D. J.; Gilliard Jr, R. J. **Stepwise reduction at magnesium and beryllium: cooperative effects of carbenes with redox non-innocent α -diimines.** *Inorganic Chemistry* **2019**, *58*, 10554–10568, **Invited Special Issue: Celebrating the Year of the Periodic Table: Emerging Investigators in Inorganic Chemistry**.

- (3) Walley, J. E.; **Obi, A. D.**; Breiner, G.; Wang, G.; Dickie, D. A.; Molino, A.; Dutton, J. L.; Wilson, D. J.; Gilliard Jr, R. J. *Cyclic (alkyl)(amino) carbene-promoted ring expansion of a carbodicarbene beryllacycle*. *Inorganic Chemistry* **2019**, 58, 11118–11126.

Patents

- (2) Deng, C.-L.; Gilliard Jr, R. J.; **Obi, A. D.**; Sarkar, S. K. Stable luminescent azaboraacenium ions, US Patent WO/2024/168273 issued Aug 15, 2024.
- (1) Gallant, B.; Kuo, F.-Y.; Junor, G.; **Obi, A. D.** Compositions, articles, and methods related to redox-active sorbents for electrochemical carbon dioxide separation, US Patent 63/737,590 filed Dec 20, 2024, 2024.

Selected Awards and Honors

CAS Future Leader, <i>American Chemical Society</i>	2024
Climate and Sustainability Impact Fellowship, <i>MIT</i>	2023
Burn Family Melville Foundation Jefferson Fellowship, <i>University of Virginia</i>	2020
Teaching Relief Fellowship, <i>University of Virginia</i>	2020
ACS Division of Inorganic Chemistry travel award, <i>American Chemical Society</i>	2020
Residency Masters in Teaching Fellowship, <i>PennGSE</i>	2015
All Academic Honors, <i>New England Small College Athletic Conference</i>	2015
Baccalaureate student speaker, <i>Bates College</i>	2015
Princeton in Africa Fellowship	2015
Sigma Xi award for academic excellence, <i>Bates College</i>	2015
STEM Research Fellowship, <i>Bates College</i>	2014
Otis Environmental Research Fellowship, <i>Bates College</i>	2013

Presentations

Oral

- (13) Gordon Research Seminar, Ventura, CA Apr 2025
A Blueprint for the Collaborative Design of Carbon Capture Materials. invited talk
- (12) University of California, Berkeley Jan 2025
Taming Electrophilic Main Group Species with Carbon Donors. invited seminar
- (11) American Chemical Society National Meeting and Expo, Denver, CO Aug 2024
A Blueprint for the Collaborative Design of Carbon Capture Materials
- (10) Case Western Reserve University, Cleveland, OH. Feb 2024
Taming Electrophilic Main Group Species with Carbon Donors. invited seminar
- (9) University of Virginia, Charlottesville, VA. Oct 2022
Electronically Flexible Carbon Donors in Magnesium and Bismuth Chemistry thesis seminar
- (8) ThermoFisher Scientific Young Chemists Webinar Series. *On Demand.* May 2022
Fundamental Bonding Situations at Magnesium and Bismuth invited seminar
- (7) American Chemical Society National Meeting and Expo, San Diego, CA Mar 2022
Carbenes and Carbones in Magnesium and Bismuth Chemistry.
- (6) Jefferson Fellows' Symposium, Charlottesville, VA. Feb 2022
Sustainable Strategies for Chemical Energy Storage.
- (5) Chemistry Seminar Series, Charlottesville, VA. Jan 2022
Electronically Flexible Carbon Donors in Magnesium and Bismuth Chemistry.

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| (4) Just Another (Chemistry) Webinar Series. Online.
<i>Dynamic NHC Coordination for the Stabilization of Organomagnesium Reagents.</i> | Dec 2021
invited talk |
| (3) German Chemical Society GDCh Phosphorus Chemistry Seminar Series. Online.
<i>A Thermally Stable Magnesium Phosphaethynolate Grignard Complex.</i> | Nov 2021
invited seminar |
| (2) PennGSE Thesis Symposium, Miss Porter's School, Farmington, CT.
<i>How can I help my students build math confidence?</i> | Apr 2017 |
| (1) Bates College, Lewiston, ME.
<i>Optical Harmonic Generation in Plasmonic Gold Nanoparticles.</i> | May 2015
thesis seminar |

Posters

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| (12) Gordon Research Conference, Ventura, CA.
<i>Templating Carbon Capture Materials design</i> | Apr 2025 |
| (11) CAS Future Leaders Symposium. Columbus, OH.
<i>A Blueprint for the Collaborative Design of Carbon Capture Materials</i> | Aug 2024 |
| (10) Gordon Research Conference. Castelldefels, Barcelona.
<i>A Blueprint for the Collaborative Design of Carbon Capture Materials</i> | Jul 2024 |
| (9) MCSC Symposium and Member Meetings. Cambridge, MA.
<i>Carbon Capture and Sequestration</i> | Nov 2023 |
| (8) Gordon Research Conference. Newport, RI.
<i>Multidimensional Carbene Coordination at Bismuth and Antimony</i> | Jun 2023 |
| (7) Gordon Research Seminar. Newport, RI.
<i>The Case for Small NHCs as Ancillary Ligands in Group 2 Chemistry.</i> | Jun 2023 |
| (6) BlackInChem Poster Competition. Online .
<i>Dynamic NHC Coordination for the Stabilization of Organomagnesium Reagents.</i> | Aug 2021
3 rd place |
| (5) UVA Chemistry Poster Symposium. Charlottesville, VA.
<i>Carbene-stabilized Organomagnesium Reagents for Bond Activation and Energy Storage.</i> | Jul 2021 |
| (4) UVA Chemistry Retreat. Crozet, VA.
<i>Carbene-stabilized Organomagnesium Reagents for Bond Activation and Energy Storage.</i> | Oct 2020
2 nd place |
| (3) UVA Chemistry Retreat. Crozet, VA.
<i>Carbene Magnesium Complexes for Small Molecule Activation.</i> | Oct 2019 |
| (2) St. Paul's School Research Symposium, Concord, NH.
<i>How can I help my students build math confidence?</i> | Apr 2017 |
| (1) Mount David Summit, Lewiston, ME.
<i>Optical Harmonic Generation in Plasmonic Gold Nanoparticles.</i> | Apr 2015 |

Teaching

Instructor on Record

Integrated Sciences II (SCI031LAB), CMC	Fall 2026
Chemistry (2 sections), Tabor	2017 - 2018
Honors Chemistry (2 sections), Tabor	2017 - 2018
Algebra II (2 sections), SPS	2015 - 2017
Algebra II Honors (2 sections), SPS	2015 - 2017
Precalculus (Math II), Phillips Andover	Summer 2017
Advanced Chemistry (Chem II), Phillips Andover	Summer 2017

Selected Topics in Chemistry (Chem 2A/2B), *Phillips Andover*

Summer 2015, 2016, 2017

Teaching Assistant

Organic Chem Lab II (Chem 2321, 4 sections), *UVA*

Spring 2019, Spring 2020

Organic Chem Lab I (Chem 2311, 4 sections), *UVA*

Fall 2018, Fall 2019

Algebra and Trigonometry (Math I), *Phillips Andover*

Summer 2016

Calculus (Math III), *Phillips Andover*

Summer 2016

Advising

Research group members

currently recruiting

Prior to Independent Career

Graduate Students: Fang-Yu Kuo (2023 – 2025, MIT), Priyanka Raghavan (2023 – 2025, MIT), Avni Singhal (2023 – 2025, MIT), Oscar Wu (2023 – 2025, MIT), Nathan Frey (2021 – 2022, UVA), Ethan SESCO (2020 – 2021, UVA), Haleigh Machost (2019 - 2022, UVA)

Undergraduate Students: Katelyn Battacharia (2025, MIT), Emiria Ibrahimovic (2025, MIT), Alexandra Chua (2024, MIT), Bishop Jackson (2024, MIT), Iraira Rivera Rojas (2024, MIT), Lucila Carlos (2024, MIT), Andrew Alexis (2021 – 2023, UVA), Samuel Coates (2020 – 2022, UVA), Eliezer Johnson (2020 - 2021, UVA)

High School Students: Caroline Bonnett (2015 – 2017, SPS), Persis Moore (2015 – 2017, SPS), Desteny Pineda (2015 – 2016, SPS), Shannice Noel (2015 – 2016, SPS), Gwen Tait (2016 – 2017, SPS), Brianna Hill (2016 – 2017, SPS), Alex Urbina (2016 - 2017, SPS), Temitope Ojo (2016 – 2017, SPS)

Selected Service and Leadership

Peer review activity

2024 - present

Nature Synthesis

Journal of the American Chemical Society

Dalton Transactions

Sustainability

Committee and board memberships

Jefferson Scholars Foundation admissions committee

2023 - present

NaijaCoder board member

2022 - present

UVA career seminar committee

2022

Tabor faculty admissions committee

2017

Other leadership positions

Co-founder and education consultant, Afternoon Prep

2018 - 2020

Treasurer, UVA Chemistry Graduate Student Council

2022

Vice president, UVA Chemistry Graduate Student Council

2021

Outreach

Science Fair Judge, NOBCCHE

2024

Invited Panelist, UVA ChemRISE seminar

2022

DEI Ambassador, UVA

2022

Youth mentor, Computers4Kids, Charlottesville, VA

2018 - 2022

Otis Fellow and Youth Coordinator, Delta Women Organization (Warri, Nigeria).

2013